

2920/206

DATABASE MANAGEMENT SYSTEM

July 2020

Time: 3 hours



THE KENYA NATIONAL EXAMINATIONS COUNCIL

DIPLOMA IN INFORMATION COMMUNICATION TECHNOLOGY

MODULE II

DATABASE MANAGEMENT SYSTEM

3 hours

INSTRUCTIONS TO CANDIDATES

*This paper consists of **EIGHT** questions.*

*Answer any **FIVE** of the **EIGHT** questions in the answer booklet provided.*

All questions carry equal marks.

Candidates should answer the questions in English

This paper consists of 5 printed pages.

Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.

1. (a) Outline **four** roles of a database administrator. (4 marks)
 - (b) Denim college has installed a database for maintaining students' records. Outline **four** benefits the college may accrue from this. (4 marks)
 - (c) Explain each of the following components of a Database Management System:
 - (i) Data Manager; (2 marks)
 - (ii) Scheduler; (2 marks)
 - (iii) Buffer Manager. (2 marks)
 - (d) With the aid of a diagram, explain the client server database architecture. (6 marks)
2. (a) Describe each of the following database application interfaces:
 - (i) Form-Based; (2 marks)
 - (ii) Graphical User. (2 marks)
 - (b) Explain **three** goals of database design phase in the database design lifecycle. (6 marks)
 - (c) Differentiate between *hierarchical* and *network* database models. (4 marks)
 - (d) A database is represented by the following schemas.

EMPLOYEE (Emp.No, Dept.No, fname, lname, salary, rank);
 DEPARTMENT (Dept.No, Dept name, Location);

Write Structured Query Language statements that could be used to create the database. (6 marks)

3. (a) Outline the function of each of the following commands:
 - (i) Insert; (2 marks)
 - (ii) Alter. (2 marks)
- (b) Distinguish between *commit* and *rollback* commands as used in Structured Query Language. (4 marks)
- (c) Johnstone, an ICT student created a database with redundant data during a practical lesson. Explain the three levels of *normalization* he could follow to eliminate this. (6 marks)
- (d) Figure 1 shows the levels of abstraction in database management systems. Explain the implementation of each level. (6 marks)

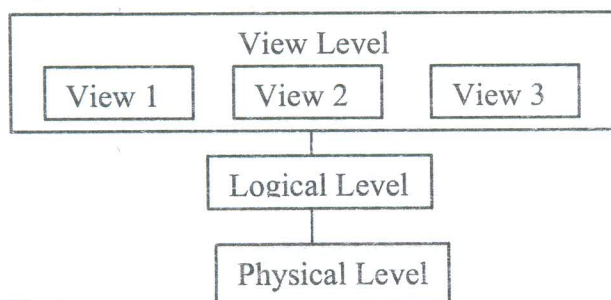


Fig.1

4. (a) Define the term *indexing* as used in Database Management Systems. (2 marks)
- (b) Distinguish between *database integrity* and *unique constraints* as used in Relational Database Management Systems. (4 marks)
- (c) Explain **three** features of a Relational Object Oriented Database Management Systems. (6 marks)
- (d) Table 1 shows a database relation named LEADERS. Use it to answer the questions that follow:

Code	Title	Allowance	TaxRate
301	CEO	200000	18
302	MANAGER	150000	17.5
303	BRANCH MANAGER	100000	16
304	ASSISTANT MANAGER	50000	15.5
305	ACCOUNTANT	25000	14.5
306	CASHIER	25000	12
307	RECEPTIONIST	12500	12

Table 1

Write Structured Query Language statements to:

- (i) Create the table *structure*; (2 marks)
- (ii) Display Code and allowance for MANAGER only; (2 marks)
- (iii) Display Title and Allowance ≥ 150000 ; (2 marks)
- (iv) Delete TaxRate attribute. (2 marks)
5. (a) Outline **four** characteristics of a database transaction. (4 marks)
- (b) Distinguish between *transitive* and *functional* dependencies as applied in Relational Database Management Systems. (4 marks)
- (c) Explain **three** causes of database transaction failure. (6 marks)
- (d) A database is represented by the following relational schemas.
 BOOK (SSID_NO, TITLE, YEAR_OF_PUBLICATION, EDITION, NUMBER_OF_PAGES);
 AUTHOR (AUTHOR NO, FNAME, LNAME, AGE);
 BORROWER (BORROWER NO, FNAME, LNAME, DATE_OF_ISSUE, DATE_OF_RETURN);
- An AUTHOR publishes one or more books
 - A BORROWER borrows one or more books
 - A BOOK with a given SSIDNO may only be borrowed by one BORROWER.
- Draw the Entity Relational diagram for the database. (6 marks)

6. (a) Outline **three** advantages of bottom-up database design approach. (3 marks)
- (b) An insurance company has implemented an Object Oriented database in its operations. Explain **three** benefits that the company may derive from the use of this database type. (6 marks)
- (c) Figure 2 shows a database organization approach. Use it to answer the questions that follow.

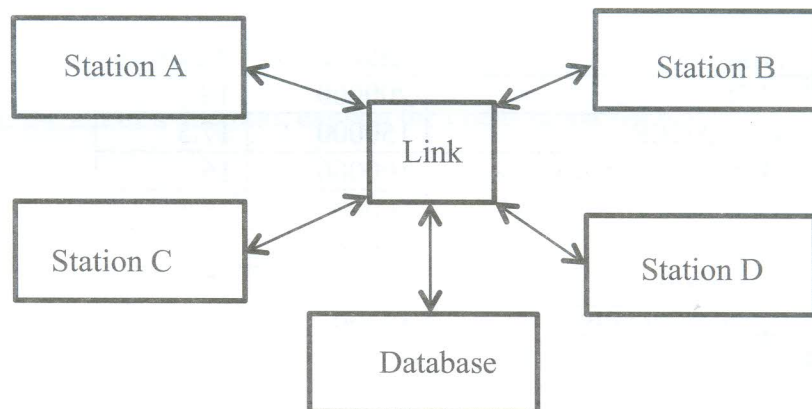


Fig. 2

- (i) Identify the approach depicted in the figure; (1 mark)
- (ii) Explain **two** benefits of the approach identified in (i). (4 marks)
- (d) With the aid of a diagram in each case, outline the type of cardinality depicted in each of the following statements:
- (i) A HOD can only manage a department. (2 marks)
- (ii) Several students enrolled in several courses. (2 marks)
- (iii) A department has several teachers. (2 marks)
7. (a) Define *tuple relational calculus* as used in databases query processing. (2 marks)
- (b) With the aid of a symbol in each case, explain the functions of each of the following relational calculus quantifiers:
- (i) Universal; (3 marks)
- (ii) Existential. (3 marks)

- (c) Tables 2 and 3 show relations in a database. Use them to answer the questions that follow.

A	B	C
A1A1	B1B1	C1C1
A2A2	B2B2	C2C2
A3A3	B3B3	C3C3

Table 2

X	B	C
X1X1	B1B1	C1C1
X2X2	B2B2	C2C2
X3X3	B3B3	C3C3

Table 3

- (i) Draw the corresponding relations for each of the following relational *algebra* operations on the tables:
- Union; (2 marks)
 - Cross-Product; (2 marks)
 - Intersection. (2 marks)
- (ii) Write equivalent structured query language statements for each of the operations in (i) (6 marks)
8. (a) Distinguish between *action query* and *Append query* as used in databases. (4 marks)
- (b) Jane a database administrator has been tasked with implementing an online database backup strategy. Outline **four** benefits of the strategy. (4 marks)
- (c) Write the SQL statements for the following relational algebra expressions.
- $\Pi_{s.name} (Supplier) \cup \Pi_{p.Color} (Product)$ (3 marks)
 - $\Pi_{Color='Red'} (Parts) \cap \Pi_{pid} (Product)$ (3 marks)
- (d) A database relation consists of the following attributes:
- ProductID, ProductName, Quantity, ComponentID, ComponentName, SupplierID, SupplierName, SupplierAddress.
- Draw the relation. (1 mark)
 - Normalize the relation in (i). (5 marks)

THIS IS THE LAST PRINTED PAGE.